AnyCompany Bank

ACB-STD-04: On-Premises Infrastructure Standards

Version 4.1 · Owner: Architecture Office · Classification: Internal (sample data)

1 Purpose and scope

This standard sets the minimum resilience requirements for systems hosted in AnyCompany Bank's own datacenters. It applies to every system with a service tier of 1, 2 or 3, and to the shared platform services those systems consume. Cloud-hosted systems are governed separately. All requirement statements use SHALL for mandatory obligations and MAY for permitted alternatives.

2 Service tiers

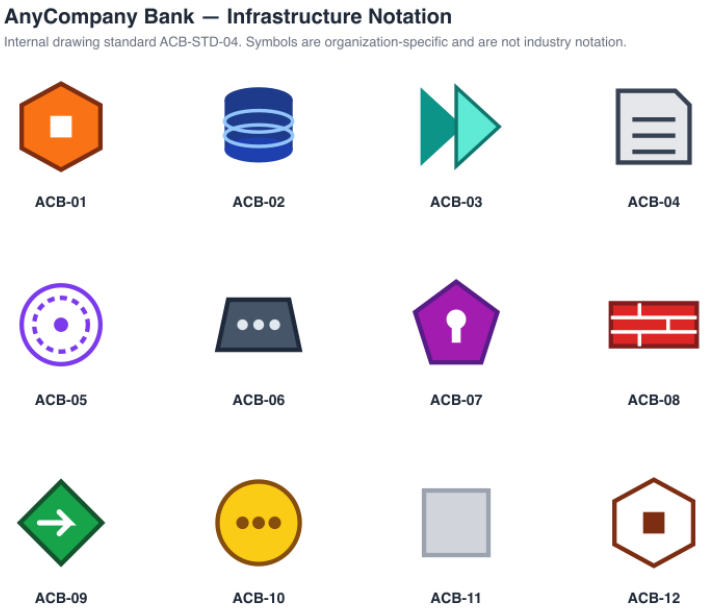
Tier	Description	Maximum outage	Data loss
Tier 1	Customer-facing payments and core ledger	15 minutes	None (RPO 0)
Tier 2	Customer-facing, non-payment services	4 hours	15 minutes
Tier 3	Internal and back-office services	1 business day	24 hours

3 Rule types

Founding rules are mandatory for the tiers they name. A **substitution** rule names an alternative that satisfies a founding rule for a given tier. An **exception** rule exempts a tier from a founding rule outright. A reviewer applies the substitution or exception before judging the founding rule failed.

4 Diagram notation

Diagrams use the house notation ACB-STD-04. The symbols are internal and are not industry icons; their meanings are published in the icon legend. For example, ACB-01 is a Tier 1 application node, ACB-02 a synchronously replicated database cluster, ACB-03 a traffic director pair, ACB-06 a SAN array, ACB-07 a hardware security module and ACB-08 a perimeter firewall.



5 Requirements

5.1 Datacenter layer

ID	Requirement	Applies to	Type	Statement	Evidence expected
ACB-REQ-001	Dual owned datacenter sites	Tier 1, Tier 2	Founding	Every Tier 1 and Tier 2 system SHALL run from two owned datacenter sites at least 30 km apart, each on independent power grids and carrier entry points.	Site list with addresses, separation distance and grid/carrier references.
ACB-REQ-002	Colocation as the second site	Tier 2	Substitution	For Tier 2 systems, a contracted colocation facility MAY substitute for the second owned site in ACB-REQ-001, provided the contract guarantees a 4-hour hands-on response.	Colocation contract reference and response-time clause.
ACB-REQ-003	Single-site exception for Tier 3	Tier 3	Exception	Tier 3 systems are exempt from ACB-REQ-001 and MAY run from a single owned site, provided nightly backups are held at the other site.	Backup schedule and off-site location.
ACB-REQ-010	Monthly site failover test	Tier 1	Founding	Tier 1 systems SHALL complete a full site failover test every month, including a return to the primary site, with results recorded in the resilience register.	Last three test dates and outcomes.
ACB-REQ-011	Quarterly failover test for Tier 2	Tier 2	Founding	Tier 2 systems SHALL complete a site failover test at least every quarter.	Last two test dates and outcomes.

5.2 Application layer

ID	Requirement	Applies to	Type	Statement	Evidence expected
ACB-REQ-004	Active-active application tier	Tier 1	Founding	Tier 1 application nodes SHALL run active-active across both sites, with no manual step required to serve traffic from either site alone.	Topology diagram in ACB-STD-04 notation showing nodes at both sites.
ACB-REQ-008	Paired hardware security modules	Tier 1, Tier 2	Founding	Hardware security modules holding payment signing keys (ACB-07) SHALL be deployed as a pair at each site, so that loss of one module or one site does not stop signing.	HSM inventory per site and key-replication method.
ACB-REQ-012	Recovery time objective	Tier 1	Founding	Tier 1 systems SHALL restore full service at the surviving site within 15 minutes of a site loss.	Measured recovery time from the most recent failover test.
ACB-REQ-014	Shared platform services are tiered	All tiers	Founding	A shared platform service (ACB-10) consumed by a system SHALL be resilient to at least the tier of its most critical consumer.	List of shared services and their tiers.

5.3 Load Balancer layer

ID	Requirement	Applies to	Type	Statement	Evidence expected
ACB-REQ-005	Traffic director pair per site	Tier 1, Tier 2	Founding	Each site SHALL front its application tier with a hardware traffic director pair (ACB-03) in active-passive mode, with health checks of 5 seconds or less.	Traffic director inventory and health-check configuration.

5.4 Data layer

ID	Requirement	Applies to	Type	Statement	Evidence expected
ACB-REQ-006	Synchronous replication for Tier 1 data	Tier 1	Founding	Tier 1 database clusters (ACB-02) SHALL replicate synchronously between sites, giving a recovery point objective of zero.	Replication mode and measured lag for each cluster.
ACB-REQ-007	Array-level replication as a substitute	Tier 2	Substitution	For Tier 2 data, asynchronous SAN array replication (ACB-06) with a recovery point objective of 15 minutes or less MAY substitute for database-level replication.	SAN replication policy and RPO evidence.

5.5 Network layer

ID	Requirement	Applies to	Type	Statement	Evidence expected
ACB-REQ-009	Segmented PCI zone	All tiers	Founding	Systems that store or process cardholder data SHALL sit inside a PCI-segmented zone (ACB-05) enforced by a perimeter firewall pair (ACB-08) at each site.	Zone diagram and firewall rule-set reference.

5.6 General layer

ID	Requirement	Applies to	Type	Statement	Evidence expected
ACB-REQ-013	Diagrams in the house notation	All tiers	Founding	Architecture diagrams submitted for review SHALL use the ACB-STD-04 notation, so that reviewers and tooling read every symbol the same way.	Diagrams with a notation reference.